



Android Application For Multi-Sensor Data Monitoring

Aniket Subham and
Shibendu Mahata

This Android-based project can monitor on-site (field) process data sensed by two temperature sensors and display the temperature (in degree centigrade) on a smartphone. The authors' prototype of the

project is shown in Fig. 1.

A smartphone-based process monitoring system is innately advantageous to personnel engaged in a wide diversity of domains. For example, the health diagnostics and subsequent

progress of a patient can be remotely monitored on the smartphone by the doctors who can then give timely guidance to the resident medical officer. Similarly, critical process variables in a process industry can also be monitored on the smartphone by the site shift engineer who can provide timely information to the field engineer to ensure process safety. Hence, a smartphone-based monitoring system, which is capable of displaying process variables from multiple sources (sensors), can prove quite beneficial.

Circuit and construction

The connections diagram for the project is shown in Fig. 2. Following components are used in this project:

LM35. This is a precision temperature sensor whose output voltage is proportional to the temperature (in °C). With an LM35, the temperature can be measured more accurately than with a thermistor. It also possesses low self-heating and does not cause more than 0.1°C temperature rise in still air due to self-heating effects. Its

operating temperature range is from -55°C to 150°C. The low output impedance, linear input-output relationship, and precise inherent calibration of LM35 make its interfacing to readout or control circuitry easy. In this project, the outputs of the LM35 sensors (SENS1 and SENS2) are connected to analogue input pins A1 and A2 of the Arduino. Power to the LM35 is provided from +5V supply in the Arduino board.

Arduino Uno. Arduino Uno is an

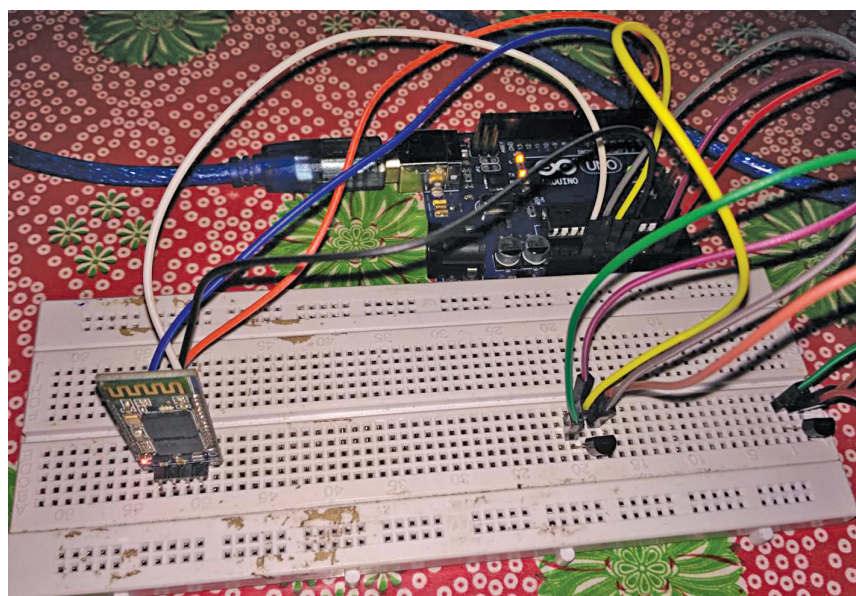


Fig. 1: Authors' prototype of the project

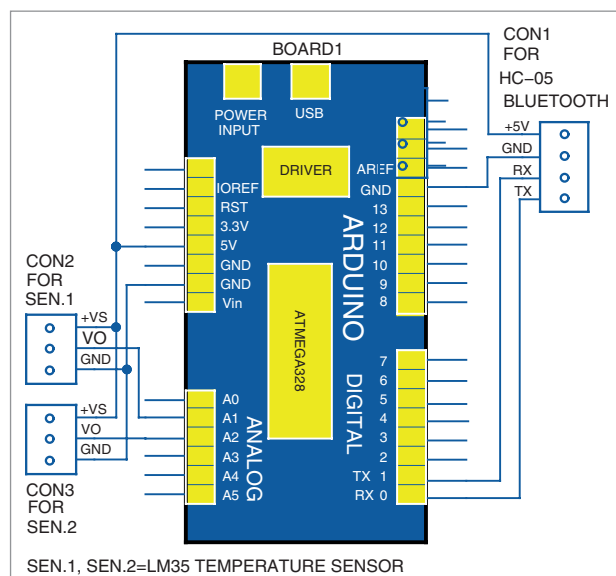


Fig. 2: Connections diagram of the project (SEN.1 and SEN.2)

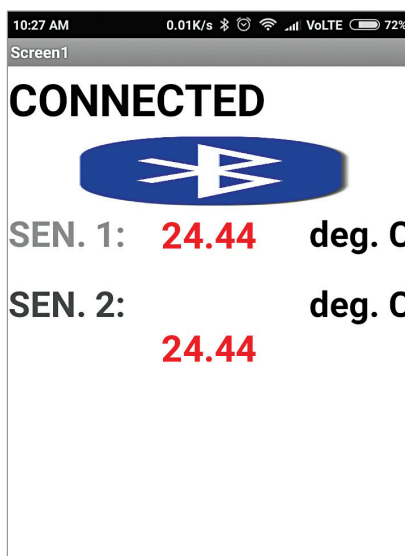


Fig. 3: Screenshot of the temperature data observed on the smartphone